

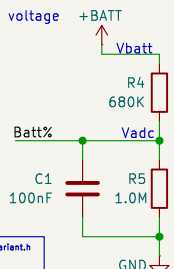
Battery voltage divider for measuring voltage

$R4 = 470$
 $R5 = 680$

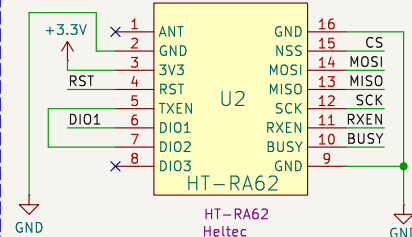
Voltage Division Equation
 $V_{adc} = V_{batt} * (R5 / (R4 + R5))$
 # Voltage $V_{batt} \rightarrow V_{adc}$ ratio
 $ADC_Division = V_{adc} / V_{batt} = (R5 / (R4 + R5))$
 $R5 / (R4 + R5) = 0.591304347826087$
 # Voltage $V_{adc} \rightarrow V_{batt}$ ratio
 $ADC_Multiplier = V_{adc} / V_{batt} = (R4 + R5) / R5$
 $(R4 + R5) / R5 = 1.6913764705882353$

C1 used for getting more stable and accurate readings

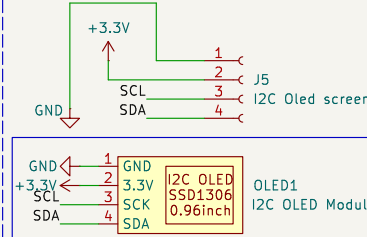
firmware/variants/nrf52840/diy/nrf52_micro_diy_tco/variant.h
 #define VBAT_DIVIDER_COMP (1.73)
 #define ADC_MULTIPLIER VBAT_DIVIDER_COMP



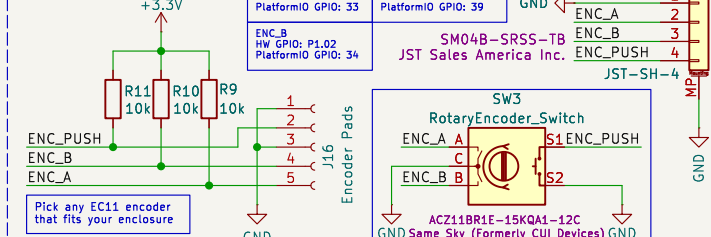
HT-RA62 / RA-01SH



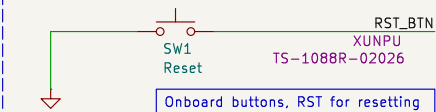
Pins for connecting an I2C OLED screen



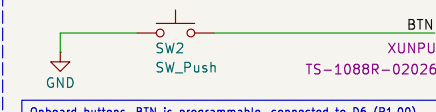
Pads for connecting encoder



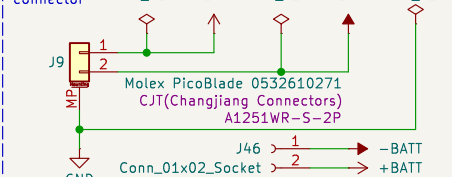
Onboard buttons RST



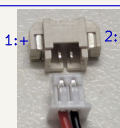
Onboard buttons BTN



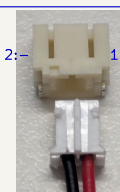
Battery connector



Molex PicoBlade 2-Pin
 - MPC: Molex, MPN: 0532610271
 - MPC: CJT (Changjiang Connectors), MPN: A1251WR-S-2P
 - Pitch: 1.25mm
 - Used By: HelTec LoRa 32
 - This polarity is also common for lipos in Aliexpress



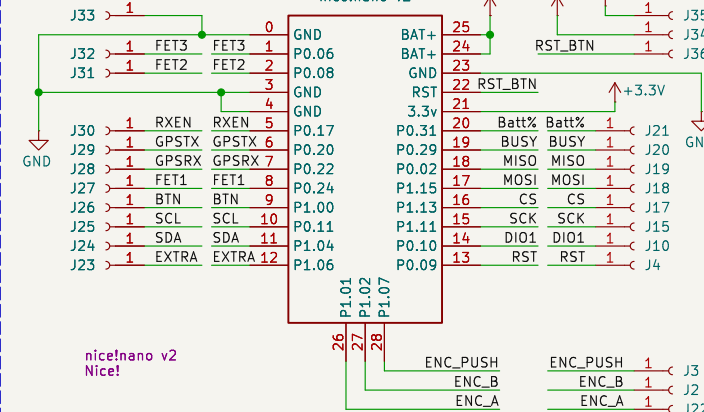
JST PH 2-Pin
 - MPN: S2B-PH-SM4-TB
 - Pitch: 2.00mm
 - Used By: RAK19007 WisBlock
 - This polarity is also common for lipos in Aliexpress



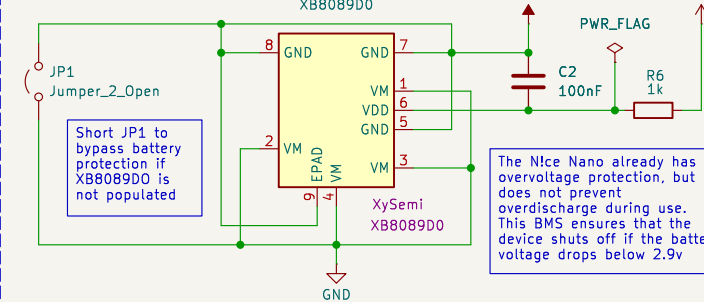
EXTRA GPIO



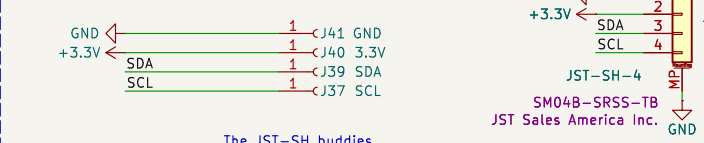
Nice!Nano nrf52840 microcontroller (or promicro)



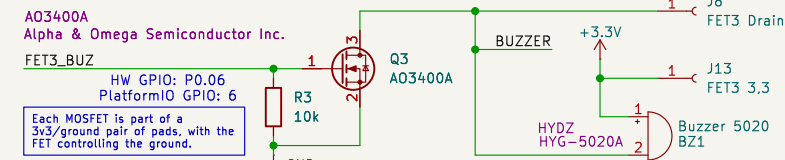
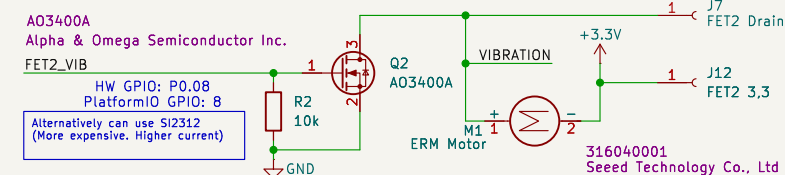
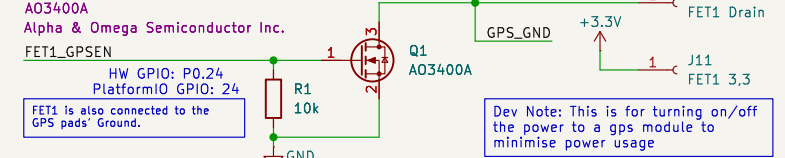
BMS for overdischarge protection



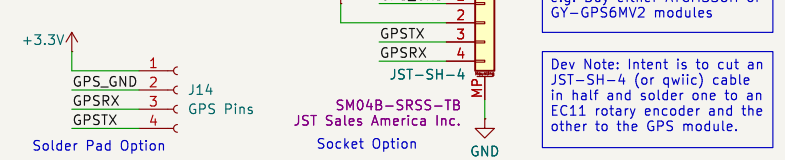
I2C Port (For extra sensors etc...)



Mosfets used for switching high current external peripheral



GPS Module



Development Source: https://github.com/mofosyne/roTec_pcb
 Fork Of FakeTec: https://github.com/jtljac/fakeTec_pcb
 Fork Of FakeTec: https://github.com/gargomoma/fakeTec_pcb/

Sheet: /
 File: fakeTecv5.kicad_sch

Title: roTec Rev5.2

Size: A4
 KiCad E.D.A. 9.0.4

Date: 2025-12-26

Rev: 5.2

Id: 1/1